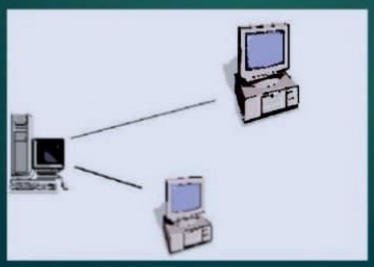
NETWORKS

Network Types

- **LAN (Local Area Network) :** A LAN (Local Area Network) is a system whereby individual PCs are connected together within a company or organization.



NETWORKS

Network Types

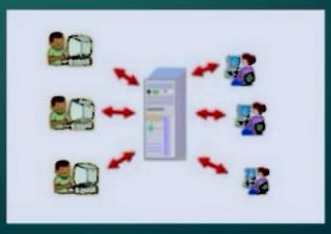
- **WLAN (Wireless Local Area Network) :** A WLAN (Wireless Local Area Network) allows you to connect to other computers within your LAN using wireless technology.

- **WAN (Wide Area Network) :** A WAN (Wide Area Network) as the name implies allows you to connect to other computers over a wider area (i.e. the whole world).

NETWORKS

Network Types

- **Client/server networks :**This term relates to the type of network where resources are kept centrally on the server and used locally by the client. The server tends to be a very Powerful PC (or group of PCs), while each client workstation, which the users have, is less powerful.



NETWORKS

Network Types

- **Internet :**The Internet is a global network of interconnected networks. The unique thing about the Internet is the sheer amount of information which you can access from it.

- **World Wide Web (WWW) vs. the Internet :**The World Wide Web (WWW) is just a small part of the Internet as a whole. The Internet relates to all the hardware and software involved, as well as the WWW, it also includes FTP (File Transfer Protocol), email and newsgroups. The WWW is basically the text and pictures which you can view using your web browser, such as Microsoft Internet Explorer, or Netscape Navigator.

NETWORKS

Network Types

- **Intranets** : An Intranet is a smaller, closed version of the Internet, which can only be accessed by authorized members of an organization. Intranets are becoming an increasingly popular way to share information within a company or other organization. An Intranet uses Internet technologies to allow users to access company documents, search databases, schedule meetings and of course send emails.

NETWORKS

Data Transfer

Downloading from & uploading to a network : You can 'download' data from a network, for instance you can download a file from a web site.

Uploading refers to copying data from your computer to another computer, either on your local area network or on your company web site.

NETWORKS

Data Transfer

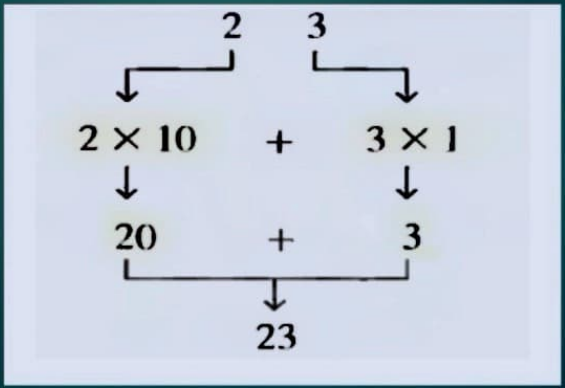
Data transfer rate measurement : When you download or upload data the speed of the data transfer is measured by the amount of data that is transferred per second. Common transfer rate units are:

- Bits per second (bps). Very slow connections will be quoted in bits per second.
- Kilobits per second (kbps).
- Megabits per second (mbps).

Numbering System

Decimal Number system

- In the decimal number system each of the ten digits, 0 through 9. represents a certain quantity.



Numbering System

Decimal Number system

- The decimal number system has a base of 10.
- The position of each digit in a decimal number indicates the magnitude of the quantity represented and can be assigned a weight. The weights for whole numbers are positive powers of ten that increase from right to left, beginning with $10^0 = 1$

$$\ldots\ 10^5\ 10^4\ 10^3\ 10^2\ 10^1\ 10^0$$

Numbering System

Decimal Number system

- For fractional numbers, the weights are negative powers of ten that decrease from left to right beginning with 10^{-1} .
- The value of a digit is determined by its position in the number. $10^2\ 10^1\ 10^0.10^{-1}\ 10^{-2}\ 10^{-3}\ldots$. The value of a decimal number is the sum of the digits after each digit has been multiplied by its weight. as Examples 1-1 and 1-2 illustrate.

Numbering System

Decimal Number system

EXAMPLE 1-1

Express the decimal number 47 as a sum of the values of each digit.

Solution

The digit 4 has a weight of 10, which is 10^1 , as indicated by its position. The digit 7 has a weight of 1, which is 10^0, as indicated by its position.

$$47 = (4 \text{ X } 10^1) + (7 \text{ X } 10^0)$$

$$= (4 \text{ X } 10) + (7 \text{ X } 1) = 40 + 7$$

Related Problem: Determine the value of each digit in 939.

Numbering System

Decimal Number system

EXAMPLE 1-2

Express the decimal number 568.23 as a sum of the values of each digit.

Solution

The whole number digit 5 has a weight of 100. which is 10^2 . the digit 6 has a weight of 10. which is 10^1. the digit 8 has a weight of 1, which is 10^0, the fractional digit 2 has a weight of 0.1, which is 10^{-1}, and the fractional digit 3 has a weight of 0.01, which is 10^{-2} .

Numbering System

Decimal Number system

$568.23 = (5 \times 10^2) + (6 \times 10^1) + (8 \times 10^0) + (2 \times 10^{-1}) + (3 \times 10^{-2})$

$= (5 \times 100) + (6 \times 10) + (8 \times 1) + (2 \times 0.1) + (3 \times 0.01)$

$500 \quad + 60 \quad + 8 \quad + 0.2 \quad + 0.03$

Related Problem : Determine the value of each digit in 67.924.

Numbering System

Binary Number System

- A binary number system is a code that uses only two basic symbols. The digits can be any two distinct characters, but it should be 0 or 1. The binary equivalent for some decimal numbers are given below

Decimal	0	1	2	3	4	5	6	7	8	9
Binary	0	1	10	11	100	101	110	111	1000	1001

Numbering System

Binary Number System

- Each digit in a binary number has a specific place value or weight based on its position. The **Least Significant Bit (LSB)** is the rightmost bit in a binary number and has a weight of **2^0=1**. The second from the right has a value of 2, the next 4 , etc.,

16	8	4	2	1
2^4	2^3	2^2	2^1	2^0

Numbering System

Binary Number System

- Binary to decimal conversion:

$(1001)2 = X_{10}$

$1001 = 1x2^3 + 0x2^2 + 0x2^1 + 1x2^0$

$= 8 + 0 + 0 + 1$

$(1001)2 = (9)_{10}$

- For fractions the weights of the digit positions are written from right of the binary point and weights are given as follows.

2^{-1}	2^{-2}	2^{-3}	2^{-4}	2^{-5}

Numbering System

Binary Number System

E.g.:

$(0.0110)_2 = X_{10}$

$= 0x2^{-1} + 1x2^{-2} + 1x2^{-3} + 0x2^{-4}$

$= 0\ x0.5 + 1x0.25 + 1x0.125 + 0x0.0625$

$= (0.375)_{10}$

E.g.:

$(1011.101)_2 = X_{10}$

$= 1x2^3 + 0x2^2 + 1x2^1 + 1x2^0 + 1x2^{-1} + 0x2^{-2} + 1x2^{-3}$

$= 8 + 0 + 2 + 1 + 0.5 + 0 + 0.125$

$= (11.625)_{10}$

Numbering System

Binary Number System

- Decimal to binary conversion:(Double Dabble method)

In this method the decimal number is divided by 2 progressively and the remainder is written after each division. Then the remainders are taken in the reverse order to form the binary number.

$(12)_{10} = X_2$

```
2 | 12
  2 | 6     - 0
    2 | 3   - 0
        1   - 1
```

$(12)_{10} = (1100)_2$

Numbering System

Binary Number System

E.g.:

$(12)_{10} = X_2$

```
2 | 21
  2 | 10     - 1
    2 | 5    - 0
      2 | 2  - 1
          1  - 0
```

$(21)_{10} = (10101)_2$

Numbering System

Binary Number System

- Fractions:
  - The fraction is multiplied by 2
  - the carry in the integer position is written after each multiplication. Then they are written in the forward order to get the corresponding binary equivalent.
  - Repeat multiplication until the fractional part becomes 0

Numbering System

Binary Number System

E.g.:

$(0.4375)_{10} = X_2$

$2 \times 0.4375 = 0.8750 \Rightarrow 0$

$2 \times 0.8750 = 1.750 \Rightarrow 1$

$2 \times 0.750 = 1.5 \Rightarrow 1$

$2 \times 0.5 = 1.0 \Rightarrow 1$

$(0.4375)_{10} = (0.0111)_2$

Binary Number System

Example Conversion of decimal 41.6875 into binary.

Numbering System

Binary Number System

Integer = 41

41	
20	1
10	0
5	0
2	1
1	0
0	1

$(41)_{10} = (101001)_2$

Fraction = 0.6875

$$
\begin{array}{r}
0.6875 \\
2 \\
\hline
1.3750 \\
\times\ 2 \\
\hline
0.7500 \\
\times\ 2 \\
\hline
1.5000 \\
\times\ 2 \\
\hline
1.0000
\end{array}
$$

$(0.6875)_{10} = (0.1011)_2$

$(41.6875)_{10} = (101001.1011)_2$

Numbering System

Binary Number System

$(0.101)_2 = (\quad)_{10}$

$(0.625)_{10} = (\quad)_2$

$(\quad)_2 = (\ 25\ )_{10}$ 11001

$(\quad)_2 = (\ 13\ )_{10}$ 1101